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1-D CNNs for structural damage detection: Verification on a structural health monitoring benchmark data

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ABSTRACT

Structural damage detection has been an interdisciplinary area of interest for various engineering fields. While the available damage detection methods have been in the process of adapting machine learning concepts, most machine learning based methods extract "hand-crafted" features which are fixed and manually selected in advance. Their performance varies significantly among various patterns of data depending on the particular structure under analysis. Convolutional neural networks (CNNs), on the other hand, can fuse and simultaneously optimize two major sets of an assessment task (feature extraction and classification) into a single learning block during the training phase. This ability not only provides an improved classification performance but also yields a superior computational efficiency. 1D CNNs have recently achieved state-of-the-art performance in vibration-based structural damage detection; however, it has been reported that the training of the CNNs requires significant amount of measurements especially in large structures. In order to overcome this limitation, this paper presents an enhanced CNN-based approach that requires only two measurement sets regardless of the size of the structure. This approach is verified using the experimental data of the Phase II benchmark problem of structural health monitoring which had been introduced by IASC-ASCE Structural Health Monitoring Task Group. As a result, it is shown that the enhanced CNN-based approach successfully estimated the actual amount of damage for the nine damage scenarios of the benchmark study.

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1. Introduction

Civil structures are susceptible to damage that can adversely affect their stiffness and stability, reducing their life-cycle performance. Numerous structural damage detection methods have been proposed in order to achieve an automated structural health monitoring (SHM) system capable of providing early warning against such structural damage [1–9]. Global (i.e. vibration-based) structural damage detection techniques have shown great promise in evaluating the condition of civil infrastructure [10,11], being categorized into parametric and nonparametric methods [12]. For damage identification, nonparametric global damage detection methods use statistical means to analyze the vibration response of the structure [13,14].

Recently, various machine learning algorithms have been used to develop more efficient parametric and nonparametric global damage detection techniques [15]. Structural damage detection methods based on machine learning essentially require two main steps: (1) feature extraction and (2) classification. In step 1, certain hand-crafted characteristics are extracted from the measured acceleration signals. Then, in step 2, a classifier is trained using the extracted features as inputs that are inherently mapped to the state of the structure being monitored, as the output.

In parametric machine learning based techniques, the modal parameters (damping ratios, mode shapes and frequencies) of the monitored structure are deduced from the vibration response and considered as extracted features [16]. On the other hand, nonparametric machine learning based methods implement other feature extraction techniques such as basic statistical analysis (utilizing the mean and variance of the signals) [17], autoregressive modeling [18], principal component analysis [19], wavelet transform [20] and other time-frequency methods [21]. For the classification step, several classifiers have been used in both parametric and

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Further back-propagation to the input delta, Δ_k^l:

$$\Delta_k^l = \frac{\partial E}{\partial y_k^l}\frac{\partial y_k^l}{\partial x_k^l} = \frac{\partial E}{\partial \mathrm{us}_k^l}\frac{\partial \mathrm{us}_k^l}{\partial y_k^l} f'\left(x_k^l\right) = \mathrm{up}\left(\Delta s_k^l\right)\beta \; f'\left(x_k^l\right) \tag{16}$$

where $\beta = (ss)^{-1}$ since each element of s_k^l is computed by averaging ss number of elements of the intermediate output, y_k^l. The inter back-propagation of the delta error ($\Delta s_k^l \overset{\Sigma}{\leftarrow} \Delta_l^{l+1}$) can be expressed as,

$$\Delta s_k^l = \sum_{i=1}^{N_{l+1}} \mathrm{conv\ 1Dz}\left(\Delta_l^{l+1},\ \mathrm{rev}\left(w_{ki}^l\right)\right) \tag{17}$$

where rev(.) is used to reverse the array and conv 1Dz(.,.) is used to perform full convolution in 1D. As such, the weight and bias sensitivities can be written:

$$\frac{\partial E}{\partial w_{ik}^l} = \mathrm{conv\ 1D}\left(s_k^l, \Delta_l^{l+1}\right) \text{ and } \frac{\partial E}{\partial b_k^l} = \sum_n \Delta_k^l(n) \tag{18}$$

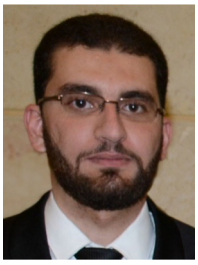
For more information on background and details, the readers are recommended to see [29,40] and [52].

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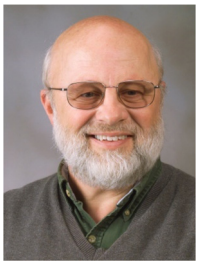
Prof. Kiranyaz published 2 books, 5 book chapters, more than 45 journal papers in 11 different IEEE Transactions and other high impact journals, and around 100 papers in international conferences. He made contributions on bio-signal analysis, particularly EEG and ECG analysis and processing, classification and segmentation, remote sensing, computer vision with applications to recognition, classification, multimedia retrieval, evolving systems and evolutionary machine learning, swarm intelligence and stochastic optimization.



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